

First-Divergence Consequence Analysis: From Quantization Onset to Perceptual Impact in Masked Visual Generators

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Abstract

Approximation error in an iterative generator is usually measured in one of two places. Local diagnostics report weight, activation, or logit distortion. Endpoint diagnostics report image quality. Neither answers the question that sits between them: at which step does the approximated sampler first behave differently, and how much does that first difference eventually cost?

We study this question for a fixed monotone masked-decoding schedule whose accepted token values are never revisited. *First-Divergence Consequence Analysis* (FDCA) runs a reference decoder and an approximate decoder from the same anchor state under an explicit coupling, defines the *first divergence* as the first transition at which their states differ, and keeps apart two quantities that endpoint scores collapse into one: how often a first split occurs, and how much terminal damage follows given that it occurred. We prove a pathwise stable-set bound, an exact onset–consequence decomposition, and a coordinate-level innovation–influence envelope under explicit product-law and hybrid-state assumptions.

We instantiate FDCA on official H-MaskGIT checkpoints with a branch-independent Halton schedule, symmetric W8 post-training weight quantization, conditional-form maximal coupling, and event-addressed random-number generation. On H-MaskGIT-T, exact split incidence *falls* from 0.213 at target anchor label $\tilde{p} = 0.3$ to 0.130 at $\tilde{p} = 0.5$, while split-conditional LPIPS *rises* by 2.98× and exact incidence-weighted LPIPS rises by 1.77×. DINOv2 distance and a matched same-shock timing control move the same way. Adding token-descendant features to a seed-only regression cuts held-out LPIPS prediction error by 48.3%. On H-MaskGIT-S the conditional and incidence-weighted ratios are 2.96× and 1.92×, and every preregistered replication test passes.

Onset and consequence therefore move in opposite directions across reverse time, making split frequency insufficient as a standalone risk measure. All claims are scoped to the audited Halton schedule, the specified coupling, a one-step natural seed followed by a common approximate suffix, and operational perceptual metrics.

Scope of this release. The empirical program is frozen. Independent artifact audits cover the primary perceptual study (H-MaskGIT-T) and the second-checkpoint replication (H-MaskGIT-S), including zero controls, semantic RNG audits, complete token and grid logging, MASK/clamp closure, and model-free reaggregation. The paper does *not* claim human perceptual ground truth, generality across architecture families, coupling-independent behavior, or any guarantee about persistent FP32-versus-W8 deployment.

1 Introduction

Post-training approximation is now a standard part of deploying a generative model. Quantization, caching, and approximate kernels all cut cost, and their error is normally checked at one of two ends of the pipeline. At the near end we measure how far the perturbed weights or logits sit from the originals. At

the far end we measure FID, a perceptual distance, or task accuracy. Neither measurement says when the approximated generator first behaved differently, how often that happens, or why two perturbations of the same local size can end up costing very different amounts.

The gap matters most in iterative masked generators. A decoder predicts tokens, commits a subset of positions, and moves on. Once one committed token differs between an exact run and an approximate run, every later conditional is evaluated on a different state, so the two runs are no longer comparable step by step. Terminal error then depends on at least three separate objects: the probability that a first split occurs at all, the geometry and severity of the token mismatch it creates, and how sensitive the remaining decoding suffix is to that mismatch. A single endpoint number hides all three. Reporting only the first — how often the runs diverge — can reverse the risk ranking, which is what we find here.

We approach the problem with a paired stochastic analysis. A reference decoder and an approximate decoder start from the same state and are joined by an explicit coupling. The *first divergence* is the first transition at which their states differ. The stopping time itself is classical: first-disagreement and maximal-agreement constructions have a long history [Lindvall, 2002, Völlering, 2016, Ernst et al., 2019]. What we add is a consequence-aware specialization to monotone masked decoding, plus an implementation contract that makes the resulting joint law auditable rather than merely asserted.

The theory has two levels. A stable-set argument shows that coordinates already fixed identically before the first split cannot contribute to terminal Hamming damage; this gives a pathwise bound that holds for any coordinate-additive cost. A time-ordered innovation–influence recursion then separates fresh same-state implementation discrepancy from the propagation of mismatches that already exist. The second level is useful as structure, but our experiments also mark its limit: finite-panel neural influence profiles order the reverse-time regimes correctly and then stop being informative inside a regime.

The empirical finding is simpler than the theory and does not depend on it. For realistic W8 weight quantization, split incidence and conditional consequence have opposite profiles across reverse time. Late transitions commit many positions at once and therefore split often, but almost no decoding remains, so the damage is small. Earlier transitions split rarely, yet the mismatch they create is amplified through a long suffix and produces much larger token and perceptual damage. The pattern survives natural quantization seeds, a matched same-shock timing control, two perceptual metrics, and a second official checkpoint.

Contributions.

1. **Theory.** We formulate FDCA for fixed monotone masked decoders and prove three results: a stable-set first-divergence bound (Theorem 4.2), an exact incidence–consequence decomposition (Proposition 4.3), and a coordinate-level clipped innovation–influence envelope (Theorem 4.4).
2. **Measurement contract.** We specify, and then audit, what a paired generative experiment has to fix in order to be reproducible: conditional-form maximal coupling, event-addressed RNG, complete token-pair and pre/post-clamp logging, semantic-key registries, zero controls, and model-free reaggregation (Section 5).
3. **Onset law under realistic quantization.** We connect W8 quantization to the mismatch seeds it actually produces and to their closed-loop propagation. Exact split incidence *decreases* toward earlier reverse-time states while conditional descendant damage *increases* sharply (Sections 7.1–7.2).
4. **Perceptual bridge.** Conditional LPIPS, exact incidence-weighted LPIPS, DINOv2 distance, and a matched timing control all show larger damage at $\tilde{\rho} = 0.5$ than at $\tilde{\rho} = 0.3$, and the primary effects replicate on a second official checkpoint (Sections 7.3–7.7).

How to read the paper. Section 3 fixes notation and defines the three damage quantities that carry most of the argument; readers who want only the empirical result can go from there to Section 7. Section 4 is self-contained and does not depend on the experiments. Section 8 lists the results that failed and the boundaries they impose.

2 Related Work

2.1 Masked visual generation and schedule design

MaskGIT introduced iterative bidirectional visual-token generation with confidence-guided refinement [Chang et al., 2022]. Later work treats the unmasking order as an algorithmic object in its own right. The Halton scheduler replaces model-dependent confidence order with a spatially dispersed low-discrepancy sequence [Halton, 1960, Besnier et al., 2025]. Jump Your Steps, factorized-approximation error bounds, and optimal inference schedules characterize the cost of parallel unmasking [Park et al., 2025, Lavenant and Zanella, 2025, Chen et al., 2026], and adaptive planning and entropy-based methods push the same point further [Kim et al., 2025, Peng et al., 2025, Ben-Hamu et al., 2025, Cai and Li, 2026]. These papers analyze sampler or scheduling bias. A complementary analysis of the MaskGIT sampler makes its implicit temperature mechanism explicit and introduces choose-then-sample and partial-caching variants [Hayakawa et al., 2025]. We hold the pretrained decoder and a branch-independent schedule fixed and study implementation approximation along paired trajectories instead.

2.2 Coupling and first disagreement

Maximal-agreement coupling and MEXIT ask how long two stochastic processes can be kept equal [Völlering, 2016, Ernst et al., 2019]; the general machinery is standard [Lindvall, 2002, Levin et al., 2017]. We use first divergence as an onset variable and claim no novelty for the stopping time. Our addition is the accounting that comes after the split, together with a contract that preserves both branch marginals and keeps the reference draw invariant across coupling families.

2.3 Local influence

Dobrushin-style comparison bounds control global discrepancy from local conditional perturbations [Rebeschini and van Handel, 2014]. Our influence matrix has the same single-site total-variation ancestry, but a fixed commit order makes the dependency graph triangular and the horizon finite. The global suprema the theorem needs are not computable for a neural decoder, so we keep the theorem and the measured reachable-state profiles strictly separate and report the latter as diagnostics.

2.4 Approximate visual inference

Caching and quantization are studied across masked, autoregressive, and diffusion-style visual generators [Yan et al., 2025, Wu et al., 2025, Moon and Ham, 2026]. Those methods optimize throughput or endpoint quality. FDCA asks a different question: where does the approximate implementation first change the stochastic trajectory, and what happens to the state mismatch afterwards?

2.5 Perceptual outcomes

We use LPIPS as a learned paired perceptual distance [Zhang et al., 2018] and DINOv2 embeddings for an independent representation-space check [Oquab et al., 2024]. A ResNet-50 classifier [He et al., 2016] supplies secondary distributional and label diagnostics. All three are operational metrics and none is a substitute for human judgment.

Table 1: Target labels and attained unresolved fractions on the frozen 16×16 , 32-step Halton schedule. Step numbers are one-indexed transition numbers.

Target $\tilde{\rho}$	Transition	Unresolved before transition	Attained ρ_{att}	Next commit batch
0.1	32	41	0.16015625	41
0.3	30	72	0.28125000	14
0.5	24	126	0.49218750	8
0.7	16	177	0.69140625	6
0.9	6	231	0.90234375	5

3 Setup and Notation

3.1 Fixed monotone masked decoder

Let I be a finite set of N visual-token positions, V a codebook, and $\mathbb{M} \notin V$ a mask symbol. A *branch-independent commit schedule* is an ordered partition

$$C = (C_1, \dots, C_K), \quad \bigsqcup_{k=1}^K C_k = I,$$

whose blocks do not depend on either branch’s realized tokens. Write $D_k = \bigcup_{j \leq k} C_j$ for the committed set and $U_k = I \setminus D_k$ for the unresolved set. A state $x_k \in (V \cup \{\mathbb{M}\})^I$ is *monotone* if a position committed to a non-mask token is never revised.

For $u \in C_{k+1}$ the reference and approximate categorical laws are $p_{k,u}(\cdot \mid x_k)$ and $q_{k,u}(\cdot \mid x_k)$. In the fixed-schedule theoretical setting, the proposal laws over a commit batch factorize,

$$P_k(\cdot \mid x_k) = \bigotimes_{u \in C_{k+1}} p_{k,u}(\cdot \mid x_k), \quad Q_k(\cdot \mid x_k) = \bigotimes_{u \in C_{k+1}} q_{k,u}(\cdot \mid x_k),$$

and coordinate couplings use independently addressed random events conditional on the common state. Both processes start from the same state and are joined by a coupling Γ . We write X_k for the reference state and Y_k for the approximate state. Factorization is needed for the product formula in Equation (1); the stable-set bound itself does not need it.

Definition 3.1 (First divergence). The first-divergence time is $\tau = \inf\{k \geq 1 : X_k \neq Y_k\}$, with $\tau = \dagger$ if the two branches never differ.

Terminal discrepancy is measured with a weighted Hamming cost

$$d_\lambda(x, y) = \sum_{u \in I} \lambda_u \mathbf{1}\{x_u \neq y_u\}, \quad \lambda_u > 0, \quad \sum_u \lambda_u = 1,$$

and the paired terminal cost is $D^\Gamma = \mathbb{E}_\Gamma[d_\lambda(X_K, Y_K)]$.

3.2 Reverse-time index and the unit of analysis

The experiment selects five target anchor labels $\tilde{\rho} \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ and then takes the nearest available *pre-transition* state in the audited 32-step Halton schedule. We write ρ_{att} for the fraction actually unresolved in that state. Large values mean *early* in decoding and small values mean *late*; “reverse time” always refers to this direction.

The distinction is material at the final anchor because the last Halton step flushes all 41 remaining positions. Table 1 gives the exact schedule-derived values. Plots use the convenient target labels $\tilde{\rho}$ on the horizontal axis; all laws and outcomes are computed at the corresponding attained state ρ_{att} .

The independent statistical unit is always the *class-seed block*: one ImageNet class label paired with one generation seed. Anchors, coupling replicates, token positions, and image pairs inside a block are not independent and are never treated as such.

Table 2: Notation used throughout.

Symbol	Meaning
I, N	set of visual-token positions; $N = I $
$C = (C_1, \dots, C_K)$	branch-independent commit schedule, K steps
$U_k, \tilde{\rho}, \rho_{\text{att}}$	unresolved set; target label; attained unresolved fraction
$p_{k,u}, q_{k,u}$	reference (FP32) and approximate (W8) conditional laws at u
X_k, Y_k	reference and approximate states
τ	first-divergence time (Definition 3.1)
d_λ	weighted terminal Hamming cost
$b_u = \text{TV}(p_u, q_u)$	per-position direct total variation ("direct TV")
$h_k(s), m_k(s)$	one-step split hazard; split-conditional terminal consequence
s_u, M_{uv}	same-state innovation at u ; single-site influence of v on u
S_{seed}	set of coordinates that differ at the first split
descendant	a coordinate that differs at the end but was not in S_{seed}
block	one class-seed pair; the independent unit of analysis

3.3 Three damage quantities, and why they differ

Most of the paper turns on keeping three numbers apart. Fix a target anchor regime $\tilde{\rho}$ (with attained state ρ_{att}) and let S be the event that a first split occurs at the anchor transition.

Quantity	Definition	Reads as
Exact split incidence	$\mathbb{P}_\Gamma(S \mid \tilde{\rho})$, exact under the specified coordinatewise coupling	How often the approximation changes behavior here
Split-conditional damage	$\mathbb{E}_\Gamma[d \mid S, \tilde{\rho}]$	How bad it is when it does
Incidence-weighted damage	$\mathbb{P}_\Gamma(S \mid \tilde{\rho}) \mathbb{E}_\Gamma[d \mid S, \tilde{\rho}]$	Expected damage attributable to a split at this anchor

Incidence is not estimated by counting splits in a Monte Carlo run. Under the coupling of Section 5.3 it is available in closed form from the per-position total variations, which removes one source of noise from every downstream contrast.

4 First-Divergence Consequence Analysis

4.1 Stable-set first-divergence bound

Definition 4.1 (Future-stable set). At a common state s immediately before transition $k + 1$, a set $F_k(s)$ is future-stable if every $u \in F_k(s)$ is equal in both branches and, under every allowed continuation of either branch, coordinate u is never revised. We suppress k when the stage is determined by the state.

Theorem 4.2 (Stable-set consequence bound). Let $d(x, y) = \sum_u \lambda_u c_u(x_u, y_u)$ be any coordinate-additive terminal cost. Then, pathwise,

$$d(X_K, Y_K) \leq \begin{cases} \sum_{u \notin F(X_{\tau-1})} \lambda_u \text{diam}(c_u) & \text{on } \{\tau \neq \dagger\}, \\ 0 & \text{on } \{\tau = \dagger\}, \end{cases}$$

and consequently $D^\Gamma \leq \mathbb{E}_\Gamma[R(X_{\tau-1}) \mathbf{1}\{\tau \neq \dagger\}]$, where $R(s) = \sum_{u \notin F(s)} \lambda_u \text{diam}(c_u)$ is the total cost mass outside the future-stable set.

For uniform token Hamming cost under the audited Halton schedule, $R(s)$ is the fraction of positions that can still be affected. The bound is exact in scope and cheap to evaluate, but it is loose whenever the first split lands early and $R(s)$ is close to one. It states what *can* be damaged, not what typically is.

4.2 Incidence and consequence

Let $G_k = \{\tau > k\}$. At a common state s , define the one-step split hazard and the conditional terminal consequence

$$h_k(s) = \mathbb{P}(X_{k+1} \neq Y_{k+1} \mid G_k, X_k = Y_k = s), \quad m_k(s) = \mathbb{E}[d_\lambda(X_K, Y_K) \mid G_k, X_k = Y_k = s, \tau = k+1].$$

When $h_k(s) = 0$, the conditional law defining $m_k(s)$ is immaterial; we adopt the convention $h_k(s)m_k(s) = 0$.

Proposition 4.3 (Exact onset–consequence decomposition).

$$D^\Gamma = \sum_{k=0}^{K-1} \mathbb{E}[\mathbf{1}\{G_k\} h_k(X_k) m_k(X_k)].$$

For a single target anchor $\tilde{\rho}$ the empirical form is $\mathbb{E}[D \mid \tilde{\rho}] = \mathbb{P}(\text{split} \mid \tilde{\rho}) \mathbb{E}[D \mid \text{split}, \tilde{\rho}]$.

The identity is elementary. Its value is entirely in what it licenses: the split law, the seed geometry, and the downstream consequence can be estimated by three separate procedures, each with its own controls, and only then multiplied.

4.3 Innovation–influence accounting

For each commit stage let S_k be an admissible state class closed under the single-coordinate hybrid substitutions used below. The same-state implementation discrepancy at a future position u is

$$s_u = \sup_{x \in S_k} \text{TV}(p_{k,u}(\cdot \mid x), q_{k,u}(\cdot \mid x)),$$

and for v committed before u the single-site influence is

$$M_{uv} = \sup_{\substack{x, y \in S_k \\ x \sim_v y}} \text{TV}(q_{k,u}(\cdot \mid x), q_{k,u}(\cdot \mid y)),$$

where $x \sim_v y$ means that x and y differ only at coordinate v . Hybrid closure is a theorem assumption; a measured collection of reachable states is not a substitute for it.

Theorem 4.4 (Clipped DAG envelope). *Order positions by commit time and define*

$$\bar{r}_u = \min\left(1, s_u + \sum_{v \prec u} M_{uv} \bar{r}_v\right).$$

Under coordinatewise maximal coupling, $\mathbb{P}(X_K[u] \neq Y_K[u]) \leq \bar{r}_u$ and $D^\Gamma \leq \lambda^\top \bar{r}$.

Because influence flows only from strictly earlier commit blocks, M is strictly block-triangular with K blocks. Hence $M^K = 0$ and

$$(I - M)^{-1} = \sum_{j=0}^{K-1} M^j$$

exists with no contraction assumption. Finiteness comes from nilpotency, not from a spectral condition.

Remark 4.5 (Measured profiles are not certificates). Theorem 4.4 is stated with suprema over all admissible states. For a neural decoder those suprema are out of reach. What we can measure are reachable-state local influences, calibration quantiles, or finite-panel maxima. Such profiles are legitimate diagnostics and we report them as such, but they do not inherit the theorem’s global language. Section 8 gives the practical cost of the gap: measured profiles order the reverse-time regimes correctly and then add almost nothing inside a regime.

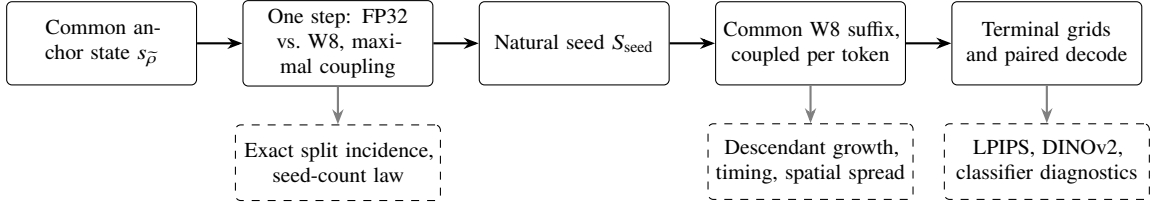


Figure 1: The natural-shock experiment. The reference and approximate kernels differ for exactly one transition. After the natural seed is created, both branches use the same W8 suffix, which isolates propagation from any further implementation innovation.

4.4 Single shock versus persistent approximation

Two estimands differ in where innovation comes from, and they must not be conflated.

Single shock with common suffix. One transition is allowed to differ, creating a seed mismatch set E . After that, both branches run the same approximate kernel. Future same-state innovation is zero, so the experiment isolates state-mediated propagation.

Persistent approximation. The reference branch keeps using p and the approximate branch keeps using q for the whole suffix. Fresh innovation continues after the first split and has to be counted. A seed-only recursion is invalid for this estimand; the finite-state audit in Appendix B includes a counterexample where omitting future innovation strictly underestimates the terminal cost.

The strongest experiments in this paper use a natural one-step seed followed by a common approximate suffix. The seed comes from the real FP32-versus-W8 one-step law, and propagation is then observed under a common W8 suffix. Everything we report about propagation is scoped to that estimand.

5 Measurement Contract

A paired stochastic experiment is reproducible only if the joint law is pinned down by the code rather than by the description. This section states the contract that was frozen and audited before any headline number was computed.

5.1 Generator and schedule

We use the official PyTorch Halton-MaskGIT implementation at source revision `f61b0a1` (full commit: `f61b0a1314717004dc7487531fd16a8bb71e1888`) [Besnier et al., 2025]. The primary checkpoint is `ImageNet_256_tiny.pth` (H-MaskGIT-T; hidden width 384, depth 6, 6 attention heads) and the replication checkpoint is `ImageNet_256_small.pth` (H-MaskGIT-S; hidden width 512, depth 12, 6 heads; 69,344,769 loaded parameters). The Small checkpoint is pinned to Hugging Face revision `ffc7fd4` (full revision: `ffc7fd4c5fc7b16010acc4aa342310484d7de62a`); its external-byte ledger records size 832,274,338 and SHA-256 `760b9937...07fb0e2`. Both configurations use a 16×16 token grid and 32 sampling steps. The Halton schedule with `randomize=False` supplies a branch-independent sequence of disjoint commit sets, and source inspection plus trajectory audits confirm that previously committed token values are never revised.

We also audited the confidence sampler and then excluded it from the exact theorem. Its *occupancy* is monotone, but committed token *values* can be rewritten. That distinction decides which sampler Theorem 4.2 applies to, and it is easy to miss.

Algorithm 1: Conditional-form maximal coupling for one token

```

Input: categorical laws  $p, q$ ; semantic keys  $k_x, k_a, k_r$ 
 $x \leftarrow \text{INVERSECDF}(p, U(k_x))$ ;
if  $\text{TV}(p, q) = 0$  then
  return  $(x, x)$ ;
 $a \leftarrow \min\{1, q(x)/p(x)\}$ ; //  $p(x) > 0$  almost surely under  $x \sim p$ 
if  $U(k_a) \leq a$  then
  return  $(x, x)$ ;
else
   $r \leftarrow (q - p)_+ / \text{TV}(p, q)$ ;
   $y \leftarrow \text{INVERSECDF}(r, U(k_r))$ ;
  return  $(x, y)$ ;

```

5.2 Approximation contract

The reference model uses FP32 weights. The approximate model applies deterministic symmetric per-output-channel round-to-nearest quantization to eligible transformer linear weights:

$$Q_b(W_{o,:}) = \text{clip}(\text{round}(W_{o,:}/\Delta_o), -q_{\max}, q_{\max}) \Delta_o, \quad \Delta_o = \frac{\max_j |W_{o,j}|}{q_{\max}}.$$

For an all-zero output row we set $Q_b(W_{o,:}) = 0$ directly. The implementation uses deterministic PyTorch round (ties to even) and zero point 0. The primary realistic cell is W8 with $q_{\max} = 127$. Activations, norms, embeddings, biases, and the VQ decoder stay in FP32. W4 is retained only as a saturated stress regime (Section 8.2).

5.3 Conditional-form maximal coupling

For categorical laws p and q , the reference token is drawn first and preserved across comparison families; the approximate token is then either accepted or drawn from the positive residual (Algorithm 1). Drawing the reference token first is not cosmetic. It keeps the reference trajectory identical across coupling variants, so a paired contrast cannot be contaminated by a change of reference draw.

The construction satisfies $X \sim p, Y \sim q$, and $\mathbb{P}(X \neq Y) = \text{TV}(p, q)$. When the commit-batch laws factorize as in Section 3 and the coordinate couplings use independent semantic events, applying the construction over a fixed batch C gives the exact split incidence under this specified coordinatewise coupling:

$$h_{\text{split}} = 1 - \prod_{u \in C} (1 - b_u), \quad b_u = \text{TV}(p_u, q_u). \quad (1)$$

This coupling is optimal for each coordinate’s mismatch probability. It is not optimal for the whole path or for any terminal image metric, and we make no such claim.

5.4 Event-addressed random-number generation

Sequential PRNG consumption silently changes the coupling whenever branch order, batching, or a rejection loop changes. Every random event is therefore addressed by a semantic key:

```

EventKey(experiment_id, model_cell, block_id, anchor_id,
         replicate_id, attempt_id, step, position, event_kind, draw_id)

```

Distinct event kinds separate reference proposals, accept/reject draws, and residual draws. Runtime registries and model-free reconstruction classify intentional reference-stream reuse and fail on any unclassified duplicate. Across the final primary studies, unclassified duplicates and missing expected keys are both zero.

5.5 State and artifact logging

The logging contract retains far more than terminal metrics: every selected token pair including equal pairs; seed and descendant coordinate vectors at each step; terminal pre-clamp and post-clamp token grids; selected-MASK and clamp-induced changes; decoded tensor, image, feature, and preprocessing hashes; source, checkpoint, and metric weight hashes; and class–seed block manifests, semantic RNG manifests, and source-to-claim tables.

A dedicated replay stage (P2B) covers more than two million selected token pairs and finds no equal-MASK and no one-sided-MASK event under the frozen P2N law. The final perceptual studies record zero selected-MASK events and zero clamp-induced changes.

6 Experimental Design

6.1 Anchors and blocks

The primary perceptual study uses target labels $\tilde{\rho} \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on 48 new class–seed blocks. The confirmatory replication uses the preregistered primary contrast $\tilde{\rho} = 0.5$ versus $\tilde{\rho} = 0.3$ on 40 power-selected new blocks. Every stage excludes all blocks used by its predecessors, so no block is reused across scientific stages.

6.2 Natural seed and common suffix

At an anchor state we compute the FP32 and W8 laws on the same next commit batch. The one-step natural seed is sampled with Algorithm 1, and conditional on a split the exact seed vector is retained, including multi-seed cases. Both branches then run the same W8 decoder over the same fixed Halton suffix. This removes further FP32-versus-W8 innovation and isolates the downstream effect of the seed, as sketched in Figure 1.

Two controls sit alongside the main panel. A matched timing panel inserts the same source-coordinate and token substitution at $\tilde{\rho} = 0.3$ and $\tilde{\rho} = 0.5$ within a block, which removes shock-identity variation between the two regimes. Zero-seed controls set both post-transition states equal and then verify exact token, image, and metric equality end to end.

6.3 Perceptual and semantic metrics

The primary metric is LPIPS with the AlexNet backbone [Zhang et al., 2018]. DINOv2 ViT-S/14 cosine distance provides a separate representation-space outcome [Oquab et al., 2024]. A pinned ResNet-50 supplies Jensen–Shannon divergence, top-1 disagreement, and conditioning-class probability and logit diagnostics. SSIM and pixel L_1 are low-level secondary measures. Metric weights, preprocessing, batch shape, and numerical tolerances were frozen before any outcome was loaded. LPIPS exceeded the scalar-versus-batch tolerance during smoke testing, so the preregistered scalar fallback was used.

6.4 Statistical tests

All headline contrasts use 20,000 paired block-bootstrap resamples over class–seed blocks. The primary perceptual hypotheses compare $\tilde{\rho} = 0.5$ with $\tilde{\rho} = 0.3$ on conditional LPIPS, exact incidence-weighted LPIPS, conditional DINOv2 distance, and matched same-shock LPIPS. A ratio threshold of 1.5 and a positive lower bootstrap endpoint were preregistered. The token-to-perceptual link is tested by leave-one-block-out ridge regression, comparing a seed-only model against a model that adds descendant count and rate, timing, and spatial spread.

Table 3: Technical audit summary for the final perceptual study and the replication.

Audit item	H-MaskGIT-T (P2P)	H-MaskGIT-S (P3S)
New scientific blocks	48; predecessor overlap 0	40; predecessor overlap 0
Zero controls	288 exact token/image/metric matches	112 exact token/image/metric matches
Selected-MASK / clamp changes	0 / 0	0 / 0
Unclassified / missing RNG keys	0 / 0	0 / 0
Complete seed-token pairs	135,514	55,314
Terminal token grids	$1,920 + 192 + 96 + 288 = 2,496$	$640 + 160 + 80 + 128 + 32 = 1,040$
Model-free reaggregation	PASS; max discrepancy 0	PASS; secondary discrepancy 0
Test inventory	60/60 frozen tests	52 pytest categories + 60 audit assertions
Package/run manifests	3,170 entries; mismatch 0	1,431 Full, 151 Summary, 1,429 run; mismatch 0

Table 4: H-MaskGIT-T under W8: exact onset law and split-conditional perceptual consequence. Incidence falls and consequence rises as the target anchor label $\tilde{\rho}$ increases.

$\tilde{\rho}$	ρ_{att}	Next batch	Split incidence	$\mathbb{E} S_{\text{seed}} $	Conditional LPIPS	Conditional DINOv2
0.1	0.1602	41	0.5008	0.6939	0.00444	0.00344
0.3	0.2813	14	0.2132	0.2382	0.01013	0.00743
0.5	0.4922	8	0.1298	0.1379	0.03013	0.02227
0.7	0.6914	6	0.1021	0.1067	0.07594	0.05634
0.9	0.9023	5	0.0833	0.0863	0.21451	0.15240

6.5 Audit chain

Table 3 summarizes the load-bearing controls. The P2P package passes a 3,170-entry run manifest with zero mismatches and 60 of 60 frozen tests. P3S passes 1,429 of 1,429 run-manifest entries, 52 pytest categories, and 60 generated audit assertions. The P3S Full-package count is 1,431 because it adds two archive-level bookkeeping files to the 1,429 run artifacts; these are not extra scientific outputs. One gap is worth stating plainly: the H-MaskGIT-S checkpoint bytes sit outside the review archive. The frozen filename, revision, size, and SHA ledger are audited, but the archive audit does not rehash the external 832 MB file.

7 Results

7.1 The onset law: many late splits, rare early splits

For H-MaskGIT-T under W8, the mean per-position direct TV is close to constant across reverse time at roughly 0.017. Split incidence nonetheless varies by a factor of six, and Equation (1) says why: the Halton schedule commits a very different number of positions per step. At the latest target anchor $\tilde{\rho} = 0.1$ (attained $\rho_{\text{att}} = 0.1602$), the next batch holds 41 positions and exact incidence is 0.501. At target $\tilde{\rho} = 0.5$ (attained $\rho_{\text{att}} = 0.4922$), it holds eight positions and incidence is 0.130. The onset law measured on the P2P blocks matches the independently generated P2N blocks closely.

7.2 Natural seeds propagate through the suffix

The natural W8 split is usually sparse. At $\tilde{\rho} = 0.3$ and $\tilde{\rho} = 0.5$, about 89% and 94% of split events contain exactly one seed coordinate, which is what the seed-count means in Table 4 imply. Conditional descendant rates are 0.0375 and 0.0848, a ratio of 2.26. These rates are normalized by the number of remaining propagation-opportunity coordinates, so the contrast is not a mechanical consequence of the earlier anchor having more unresolved positions. At $\tilde{\rho} = 0.5$, 97.4% of natural split replicates create at least one descendant mismatch, so extinction of the seed is the exception rather than the rule. The

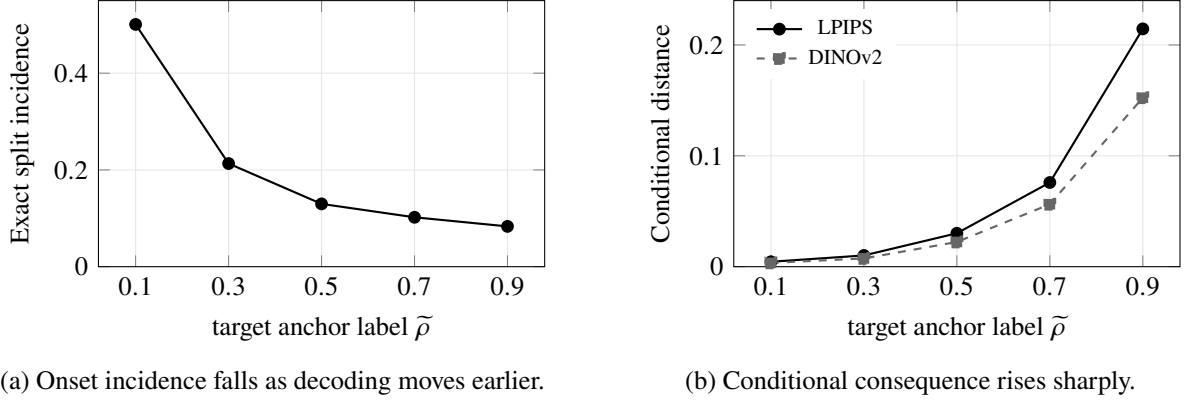


Figure 2: Opposing reverse-time profiles for H-MaskGIT-T under W8. The horizontal axis uses target labels $\tilde{\rho}$; attained fractions are listed in Table 4. Splits are most frequent late in decoding, where little suffix remains; conditional perceptual damage is largest early, where a great deal remains.

Table 5: Primary H-MaskGIT-T perceptual results, target $\tilde{\rho} = 0.5$ versus $\tilde{\rho} = 0.3$. Intervals are 95% paired block-bootstrap intervals over 48 class-seed blocks.

Outcome	$\tilde{\rho} = 0.5$	$\tilde{\rho} = 0.3$	Ratio	Difference	95% CI for difference
Conditional LPIPS	0.03013	0.01013	2.975	0.02000	[0.01719, 0.02287]
Incidence-weighted LPIPS	0.003927	0.002213	1.775	0.001714	[0.001368, 0.002066]
Conditional DINOv2	0.02227	0.00743	2.996	0.01484	[0.01099, 0.01878]
Matched same-shock LPIPS	0.04004	0.01495	2.678	0.02509	[0.01552, 0.03588]

matched same-shock timing control yields a comparable twofold token-level contrast, which separates the effect of reverse-time position from the identity of the shock.

7.3 Perceptual consequence on H-MaskGIT-T

Table 5 reports the preregistered P2P hypotheses. All four pass the $1.5\times$ ratio threshold with positive lower bootstrap endpoints.

The conditional LPIPS contrast is positive in 47 of the 48 blocks. It stays positive under block medians, under restriction to single-seed events, after excluding extinctions, after removing the top 1% of rows, and after adjusting for seed severity. DINOv2 gives an almost identical reverse-time ratio despite being a different metric on a different backbone. In the matched timing panel, where the source coordinate and the alternative token are held fixed inside each block, the $\tilde{\rho} = 0.5$ damage is still $2.68\times$ larger.

7.4 Incidence offsets consequence without cancelling it

Exact split incidence is lower at $\tilde{\rho} = 0.5$ than at $\tilde{\rho} = 0.3$, so the two effects push against each other in the unconditional metric. Writing $\bar{h}$ and $\bar{m}$ for the block-aggregated incidence and conditional damage, a symmetric two-factor attribution of the unconditional LPIPS difference is

$$\Delta_{\text{incidence}} = (\bar{h}_{.5} - \bar{h}_{.3}) \frac{\bar{m}_{.5} + \bar{m}_{.3}}{2}, \quad \Delta_{\text{consequence}} = (\bar{m}_{.5} - \bar{m}_{.3}) \frac{\bar{h}_{.5} + \bar{h}_{.3}}{2},$$

which evaluates to

$$\Delta_{\text{incidence}} = -0.001743, \quad \Delta_{\text{consequence}} = +0.003457, \quad \Delta_{\text{total}} = +0.001714.$$

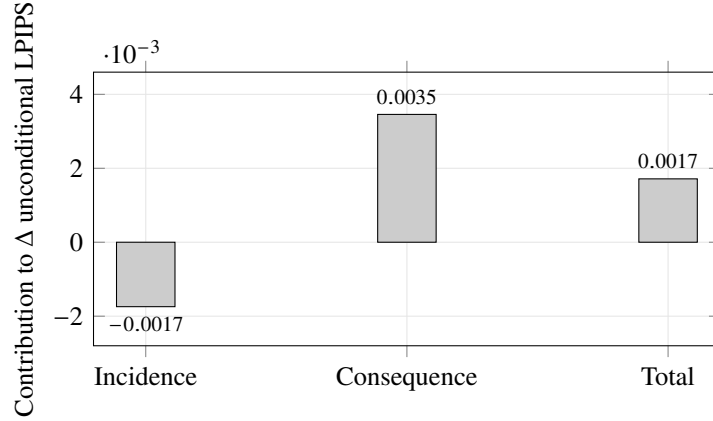


Figure 3: Opposing contributions to the $\tilde{\rho} = 0.5$ versus $\tilde{\rho} = 0.3$ unconditional LPIPS contrast. Lower incidence at $\tilde{\rho} = 0.5$ subtracts from the difference; the larger conditional consequence more than makes up for it.

The incidence term removes about half of the consequence term, and what is left is still positive.¹ The attribution is descriptive rather than unique: a sequential decomposition assigns the interaction term differently, and we read no causal weight into either factor.

7.5 Token propagation predicts perceptual damage

A leave-one-block-out seed-only model uses the target-anchor category, seed count, direct-TV severity, surprisal, and seed topology. Adding descendant count and rate, first descendant step, and spatial spread reduces held-out LPIPS mean absolute error from 0.03035 to 0.01568, a ratio of 0.5166. The block-bootstrap interval for the improvement is [0.01240, 0.01702], and the within-anchor residual Spearman correlation is 0.368 with interval [0.313, 0.423].

Descendant count or rate carries almost all of the incremental information. Spatial spread on its own is weaker, and adding spread after descendant count moves MAE by less than one percent. This is a predictive association measured after the suffix has already unfolded. It is not an early-warning model, and Section 9.2 says what would have to change for it to become one.

7.6 Semantic and low-level corroboration

Classifier Jensen–Shannon divergence and the absolute change in conditioning-class probability and logit both increase from $\tilde{\rho} = 0.3$ to $\tilde{\rho} = 0.5$. Conditional top-1 disagreement rises from 1.56% to 4.43%, though the event count behind that comparison is sparse. The *signed* conditioning-class probability drop crosses zero, so the data support a distributional perturbation rather than a consistent loss of the intended class. Pixel L_1 and $1 - \text{SSIM}$ move in the same reverse-time direction.

7.7 Second-checkpoint replication

The minimal confirmatory study swaps H-MaskGIT-T for the official H-MaskGIT-S checkpoint and keeps everything else fixed. A preregistered power calculation selected 40 new blocks. All four primary tests replicate (Table 6).

The conditional LPIPS difference retains 101.9% of the H-MaskGIT-T effect and the incidence-weighted difference retains 114.7%. The DINOv2 absolute difference is smaller on the larger model, but its

¹Both terms are computed from per-block aggregates in the frozen analysis, not from the rounded anchor means printed in Table 4. Recomputing from those rounded means shifts the individual terms in the third decimal place and leaves the signs and the total unchanged.

Table 6: Second-checkpoint replication on H-MaskGIT-S, target $\tilde{\rho} = 0.5$ versus $\tilde{\rho} = 0.3$, over 40 new class–seed blocks.

Outcome	$\tilde{\rho} = 0.5$	$\tilde{\rho} = 0.3$	Ratio	Difference	95% CI
Conditional LPIPS	0.03076	0.01038	2.963	0.02038	[0.01714, 0.02381]
Incidence-weighted LPIPS	0.004106	0.002140	1.919	0.001966	[0.001499, 0.002477]
Conditional DINOv2	0.01399	0.00457	3.062	0.009421	[0.007297, 0.011919]
Matched same-shock LPIPS	0.03227	0.01004	3.215	0.02223	[0.01603, 0.02886]

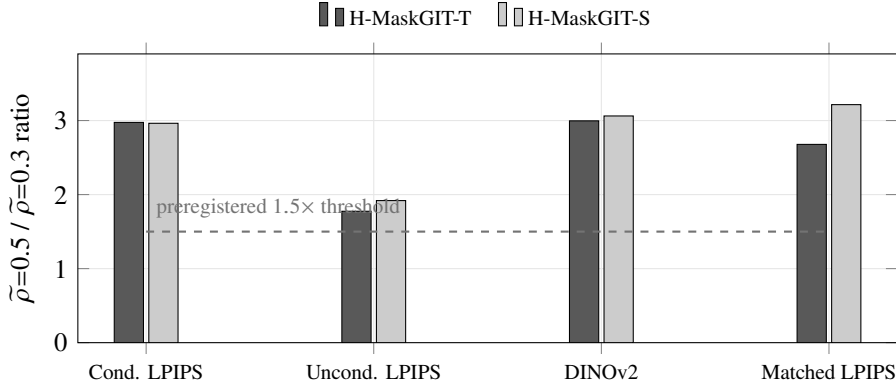


Figure 4: Primary reverse-time ratios on two official checkpoints from the same H-MaskGIT family. The dashed line marks the preregistered $1.5\times$ threshold.

reverse-time ratio stays near three. The token-to-perceptual link also replicates: seed-only and propagation-model held-out MAE are 0.01280 and 0.006964 (ratio 0.544), the block-level MAE improvement is 0.005838 with interval [0.004914, 0.006790], and the within-regime Spearman correlation is 0.584 with interval [0.501, 0.664]. A 16-block official-CFG sensitivity panel points the same way and remains secondary.

8 What the Data Do Not Support

Several results constrain the claim, and reporting them is part of the claim.

8.1 Measured influence profiles are only a coarse ordering

Reachable-state q95 and calibration-maximum profiles achieve high finite-panel coverage, but the clipped envelope they produce saturates in the early, high- $\tilde{\rho}$ regimes where a bound would be most useful. Adding the static profile to a structural baseline gives essentially no out-of-fold MAE improvement within a fixed ρ . The profile works as a regime ordering and as a conservative diagnostic, and it does not work as a certificate; Remark 4.5 anticipated the reason.

8.2 W4 is saturated

W4 direct TV is about an order of magnitude larger than W8, split incidence is near one, and multi-seed events dominate. W4 confirms the direction of the mechanism under a severe stress test, but the regime is far from the deployment setting this paper is about, so we do not treat it as primary evidence.

8.3 Source age is confounded

Earlier-committed sources are associated with larger propagation under the frozen selection rule. The association becomes unstable once source and shock characteristics are adjusted for, so we make no causal

claim about source age.

8.4 No phase transition is established

Damage increases sharply with the target anchor label $\tilde{\rho}$, but we neither fit nor test a discontinuous changepoint. The correct term for what we measured is *reverse-time-dependent amplification*.

9 Discussion

9.1 Why split frequency fails as a standalone risk metric

A deployment monitor that reports only the probability of divergence will rank the last decoding steps as the most dangerous, because those steps commit the most positions at once. On our data that ranking is backwards for expected consequence. The same approximate implementation has near-constant per-position direct TV across reverse time, yet suffix sensitivity changes by more than an order of magnitude. Expected damage is a product of onset and consequence, and Section 7.4 shows the two factors can carry opposite signs.

9.2 What the instrumentation would enable next

The experiments here use full suffix information, which makes them diagnostic rather than a controller. What the instrumentation does provide is the separated set of quantities a controller would need: the exact local split law, seed count and severity, first descendant time, descendant growth, and final perceptual consequence. A cost-saving system built on this should be held to a stricter target than anything we test. Can *pre-terminal* features identify the small subset of transitions that deserve higher precision or recomputation? This paper establishes the mechanism and the measurement contract and leaves that question open.

9.3 Where the theory and the experiments meet

The stable-set theorem is the robust part. It holds at the architecture level and needs only monotonicity and a branch-independent schedule. The innovation–influence recursion shows how a finer bound could be built if global local-sensitivity moduli were available, which for a neural decoder they are not. The experiments then locate the useful interface: first-divergence onset and conditional consequence carry the argument, while static influence profiles stay secondary. That narrowing is itself a result. The closed-loop descendant count turned out to be far more predictive than any static profile we measured.

10 Limitations

- The two checkpoints belong to the same H-MaskGIT family, share the VQ decoder, and use the same Halton scheduler. The results establish robustness across checkpoint and scale, not generality across architecture families.
- The joint law is defined by a specified coordinatewise conditional maximal coupling. A different valid coupling can change paired path costs.
- The natural-seed experiment is not persistent FP32-versus-W8 deployment. After one FP32-versus-W8 transition, both branches use a common approximate suffix (Section 4.4).
- LPIPS, DINOv2, and the classifier diagnostics are operational metrics. No human perceptual study is included.
- The primary mechanism uses Halton with `randomize=False`. The audited confidence sampler revises committed token values and would need a different multitype theory.

- We establish no global or tight neural influence certificate, no FID improvement, no caching benefit, and no production safety guarantee.
- The H-MaskGIT-S checkpoint bytes are external to the review archives. Independent audit verifies the frozen ledger, revision, size, and SHA record, but does not rehash the excluded 832 MB file.

11 Conclusion

FDCA separates the onset of approximation-induced behavioral change from the consequence that follows it. In masked visual generation the separation is not cosmetic: W8 split incidence is highest late in decoding, while token and perceptual damage conditional on a split is several times larger in earlier states. Natural quantization seeds, common-suffix propagation, two paired perceptual metrics, a matched same-shock control, and a second official checkpoint all point the same way. For anyone monitoring an approximated iterative generator, the practical consequence is that a divergence counter is the wrong instrument on its own. Expected damage is the product of a split law, a seed geometry, and the amount of generator state still to be decided.

Data, Code, and Version History

Release. This document is version 1.0.0, the first public archival release of the work. The venue-neutral layout and non-anonymous author metadata are intentional for this technical-preprint record. Earlier numbered manuscripts were internal development drafts and were not published as Zenodo records, releases, or citable public versions.

Initial-public-release review. Before this release, the manuscript underwent a final correctness, scope, provenance, and artifact audit. The review distinguished target anchor labels from attained unresolved fractions; stated the batch-factorization and hybrid-closure assumptions; added zero-hazard and zero-weight-row conventions; hardened Algorithm 1 at $TV(p, q) = 0$; recorded exact upstream and checkpoint configurations; completed the H-MaskGIT-S prediction table; reconciled terminal-grid and package-manifest arithmetic; incorporated directly relevant MaskGIT sampler work; and aligned the paper, citation metadata, license, and public manifests. No frozen experimental estimate or scientific gate was changed by this release-engineering pass.

Artifacts. The initial public release contains the PDF, LaTeX and BibTeX sources, machine-readable citation and Zenodo metadata, license, release notes, and compact independent-audit reports. Large trajectory archives, checkpoint weights, and metric weights are not redistributed here. The paper and audit reports identify filenames, revisions, sizes, and SHA-256 ledgers.

A Proofs

A.1 Proof of Theorem 4.2

Proof. On $\{\tau = \dagger\}$ the two branches coincide at every step, so $d(X_K, Y_K) = 0$ and the bound holds trivially.

On $\{\tau \neq \dagger\}$ the two states are equal immediately before the first split. Every coordinate in $F(X_{\tau-1})$ is equal at that time and is never revised in either branch, so its terminal cost is zero. All terminal discrepancy is therefore confined to the complement of $F(X_{\tau-1})$, and each surviving coordinate cost is bounded by $\text{diam}(c_u)$. Summing over $u \notin F(X_{\tau-1})$ gives the pathwise bound, and taking expectations gives the bound on D^Γ . $\square$

A.2 Proof of Proposition 4.3

Proof. The events $\{\tau = k + 1\}$ for $k = 0, \dots, K - 1$ partition $\{\tau \neq \dagger\}$, and $d_\lambda(X_K, Y_K) = 0$ on $\{\tau = \dagger\}$. Hence

$$D^\Gamma = \sum_{k=0}^{K-1} \mathbb{E}[d_\lambda(X_K, Y_K) \mathbf{1}\{\tau = k + 1\}].$$

Condition on G_k and on the common state $X_k = Y_k$. The conditional probability of $\{\tau = k + 1\}$ is $h_k(X_k)$ by definition, and the conditional terminal cost on that event is $m_k(X_k)$. Applying the tower property term by term gives the claim. $\square$

A.3 Proof of Theorem 4.4

Proof. Proceed in commit order and write $\pi_u = \mathbb{P}(X_K[u] \neq Y_K[u])$. When u is committed, coordinate-wise maximal coupling makes its mismatch probability equal to $\text{TV}(p_u(\cdot | X), q_u(\cdot | Y))$. Insert the intermediate law $q_u(\cdot | X)$ and apply the triangle inequality for total variation. The same-state term is at most s_u by definition. For the remaining term, move from X to Y along a hybrid path that changes one previously committed mismatched coordinate at a time; each single-coordinate change contributes at most M_{uv} and occurs with probability at most π_v , so the accumulated state term is at most $\sum_{v \prec u} M_{uv} \pi_v$. Therefore

$$\pi_u \leq \min\left(1, s_u + \sum_{v \prec u} M_{uv} \pi_v\right).$$

The recursion defining $\bar{r}$ is monotone in its inputs, so induction over the commit order gives $\pi_u \leq \bar{r}_u$ for every u , and the weighted terminal bound $D^\Gamma \leq \lambda^\top \bar{r}$ follows by summation.

A position is influenced only by positions in strictly earlier commit blocks, so M is strictly block-triangular with K blocks. Hence $M^K = 0$ and $(I - M)^{-1} = \sum_{j=0}^{K-1} M^j$, which is where the unclipped version of the envelope comes from. No contraction assumption is used anywhere. $\square$

B Experimental Program

Table 7: Stages that produced the final claim, in order.

Stage	Purpose	Outcome
T0	Exact finite-state theorem and code validation	Stable-set bound, clipped envelope, Wasserstein ordering, and the fresh-innovation counterexample all pass.
P0	Official sampler contract	Halton is branch-independent and token-value monotone; the confidence sampler is monotone in occupancy only.
P05R	Real-model influence microaudit	Profiles tighten the late regimes but fail uniform global criteria; the profile is demoted to a secondary diagnostic.
P1L	Controlled single shock	Strong reverse-time token amplification and spatial spread.
P1M	Model-free incremental analysis	Profile adds little within-anchor value; the regime-level result is robust.
P2N	Natural W8 seed bridge	Exact incidence, natural seed, conditional propagation, and matched timing all supported.
P2A/P2B	RNG, MASK, and clamp closure	Unclassified key duplicates 0; hidden MASK and clamp artifacts excluded.
P2P	Perceptual bridge	LPIPS, DINOv2, token link, and matched timing (H1–H5) all pass.
P3S	Second checkpoint	R1–R4 and the token link replicate on H-MaskGIT-S.

The T0 counterexample referenced in Section 4.4 is a finite-state chain on which a seed-only recursion, applied to the persistent-approximation estimand, strictly underestimates the terminal cost. It is the reason the two estimands are kept apart throughout the paper.

C Full H-MaskGIT-T Perceptual Tables

Table 8 gives the conditional and exact incidence-weighted outcomes at every anchor. Conditional columns answer "how bad if it splits"; unconditional columns are the conditional value multiplied by the exact incidence of the same row, following Proposition 4.3. Reading across a row shows the two directions cancelling; reading down the unconditional LPIPS column shows that the cancellation is incomplete.

Table 8: Conditional and exact incidence-weighted perceptual outcomes by reverse-time anchor (H-MaskGIT-T, W8, 48 blocks).

$\tilde{\rho}$	ρ_{att}	Split incidence	Cond. LPIPS	Uncond. LPIPS	Cond. DINOv2	Uncond. DINOv2	Cond. JSD
0.1	0.1602	0.50079	0.004441	0.002267	0.003435	0.001699	0.000611
0.3	0.2813	0.21323	0.010126	0.002213	0.007432	0.001512	0.001558
0.5	0.4922	0.12976	0.030126	0.003927	0.022270	0.002830	0.004861
0.7	0.6914	0.10207	0.075937	0.007869	0.056340	0.005639	0.010492
0.9	0.9023	0.08329	0.214513	0.017798	0.152405	0.012133	0.024952

Table 9: Seed geometry and propagation at the two preregistered anchors (H-MaskGIT-T, W8).

Quantity	$\tilde{\rho} = 0.3$	$\tilde{\rho} = 0.5$
Positions in next commit batch	14	8
Exact split incidence	0.2132	0.1298
Expected seed count $\mathbb{E} S_{\text{seed}} $	0.2382	0.1379
Share of splits with exactly one seed coordinate	$\approx 89\%$	$\approx 94\%$
Conditional descendant rate	0.0375	0.0848

Table 10: Leave-one-block-out prediction of terminal LPIPS. The seed-only model uses the target-anchor category, seed count, direct-TV severity, surprisal, and seed topology; the propagation model adds descendant count and rate, first descendant step, and spatial spread. The replication entries come from the frozen P3S out-of-fold audit.

Quantity	H-MaskGIT-T	H-MaskGIT-S
Seed-only held-out MAE	0.03035	0.01280
Propagation held-out MAE	0.01568	0.006964
MAE ratio	0.5166	0.5440
Bootstrap interval for the MAE improvement	[0.01240, 0.01702]	[0.004914, 0.006790]
Within-regime residual Spearman	0.368 [0.313, 0.423]	0.584 [0.501, 0.664]

D Reproducibility and Audit Notes

Block independence. Every scientific stage uses deterministic class–seed manifests and excludes all predecessor blocks. The primary perceptual study uses 48 new blocks; the replication uses 40 new blocks chosen by a preregistered power calculation.

Zero controls. Zero-seed controls produce identical pre-clamp and post-clamp token grids, identical decoded image hashes, and exactly zero LPIPS, DINOv2, and classifier distance. These controls exercise the whole coupling–decode–metric pipeline rather than the token sampler alone, which is why they are run at the same scale as the scientific panels.

MASK and clamp. A targeted replay retains every equal and mismatched token pair at the natural seed transition. Across more than two million selected token pairs, no equal-MASK and no one-sided-MASK event occurs. The perceptual panels record zero selected-MASK events and zero clamp-induced changes.

RNG. The semantic key registry classifies intentional cross-cell reuse of the reference proposal stream and fails on any unclassified duplicate. The final primary and replication panels contain zero unclassified, missing, or unexpected keys.

Artifact integrity. Run manifests are non-self-referential and carry size and SHA entries. The P2P independent analysis clean-extracts the Full package and reproduces the gate. Its 2,496 terminal grids decompose as 1,920 W8 natural pairs, 192 W4 stress pairs, 96 matched-timing pairs, and 288 zero controls. The P3S 1,040 grids decompose as 640 primary natural pairs, 160 matched timing pairs, 80 primary zero controls, 128 official-CFG sensitivity pairs, and 32 sensitivity zero controls. The P3S independent audit verifies the Full, Summary, and run manifests, all headline arithmetic, the zero controls, the logging rows, the source code, and a secondary implementation. Its Full package contains the 1,429 run-manifest files plus two archive-level bookkeeping files.

E Claim–Evidence Matrix

Table 11: Every claim in the paper with its evidential status.

Claim	Status	Evidence and qualification
Stable-set first-divergence bound	Theorem	Fixed monotone common schedule.
Coordinate clipped envelope	Theorem	Requires global admissible-state innovation and influence moduli.
Measured influence profile is tight	Not supported	Conservative, saturates early, weak within-anchor prediction.
Natural W8 seed propagates	Supported	P2N; common W8 suffix, specified coupling.
Reverse-time token amplification	Supported	Controlled shock, natural seed, matched timing.
Perceptual reverse-time amplification	Supported	LPIPS and DINOv2 on H-MaskGIT-T and H-MaskGIT-S.
Token propagation predicts LPIPS	Supported	Leave-one-block-out improvement on both checkpoints; not an online early-warning model.
Source age causes propagation	Not supported	Association is confounded and unstable under adjustment.
Human perceptual failure	Untested	Operational metrics only.
Persistent deployment guarantee	Untested	One-step natural seed plus common suffix.
Architecture-family generality	Untested	Two checkpoints in one H-MaskGIT family.
Coupling-universal result	Untested	Coordinatewise conditional maximal coupling.

F Terminology Checklist

The following distinctions are load-bearing and are applied consistently above.

- Write *reverse-time-dependent amplification*, not *phase transition*.
- Write *operational perceptual distance*, not *human perceptual failure*.
- Name the specified coupling whenever a paired damage number is reported.
- Keep *exact split incidence under the specified coordinatewise coupling*, *split-conditional consequence*, and *incidence-weighted consequence* separate (Section 3.3).

- Keep the *natural one-step seed plus common suffix* estimand separate from *persistent approximation* (Section 4.4).
- Report measured influence profiles as finite-panel diagnostics, never as global certificates (Remark 4.5).
- Keep W8 as the primary realistic regime and W4 as a saturated stress panel.
- Distinguish target anchor label $\tilde{\rho}$ from attained unresolved fraction ρ_{att} .

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